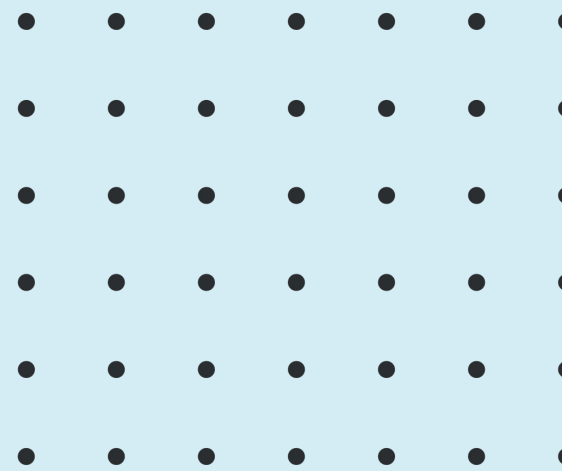


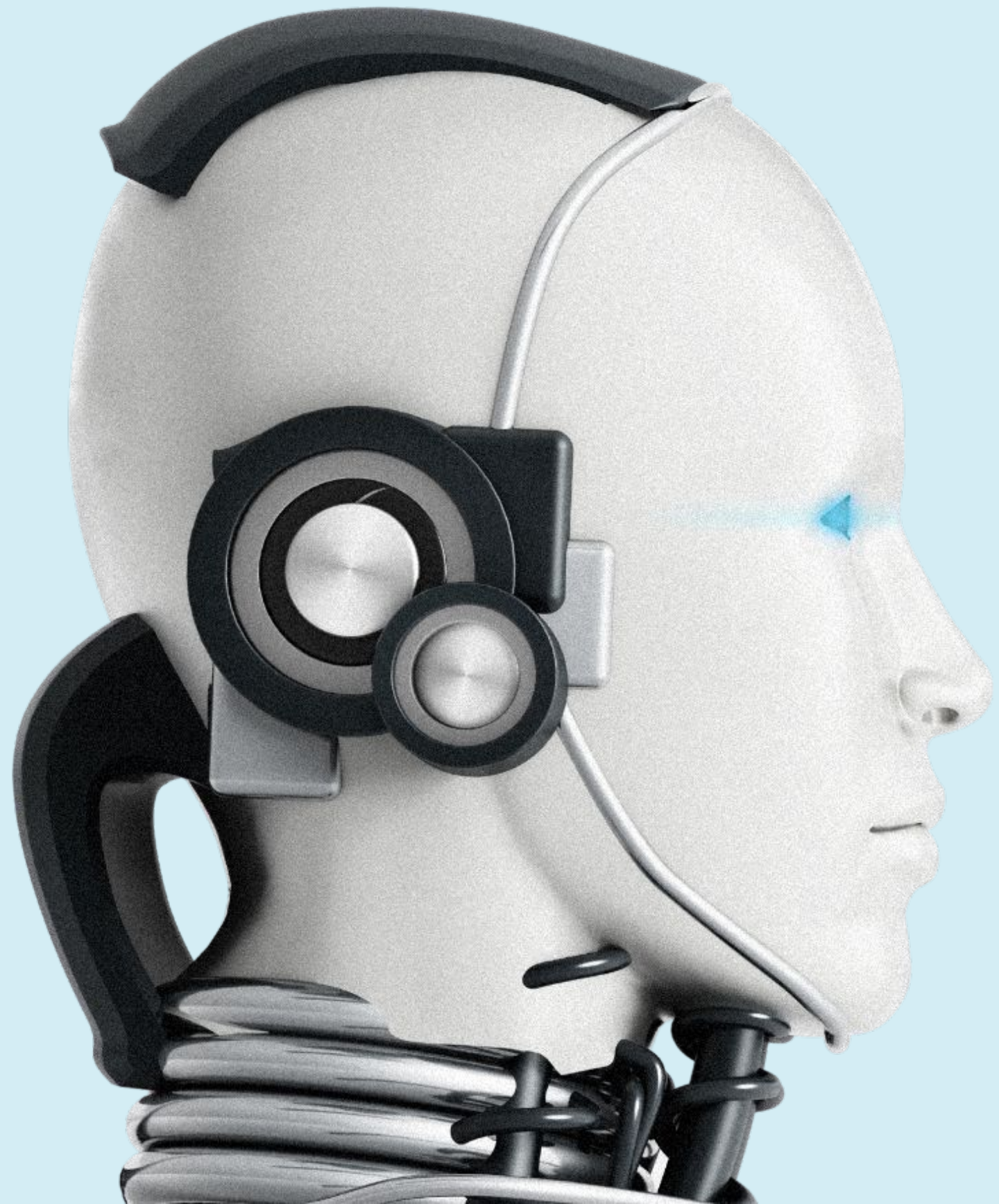
AI

Smart Face Attendance System



Introduction

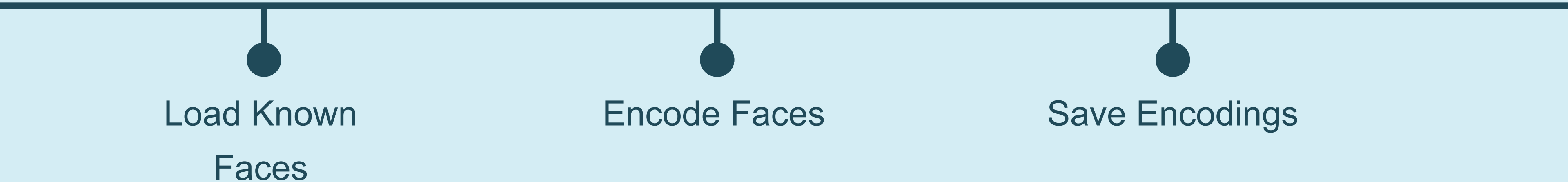
- Traditional attendance is slow, manual, and prone to errors.
- Smart campuses require automated and contactless solutions.
- This system uses face recognition to
detect and record student attendance in
real time.



AI

System Overview

- **Preparation (Face Encoding)**



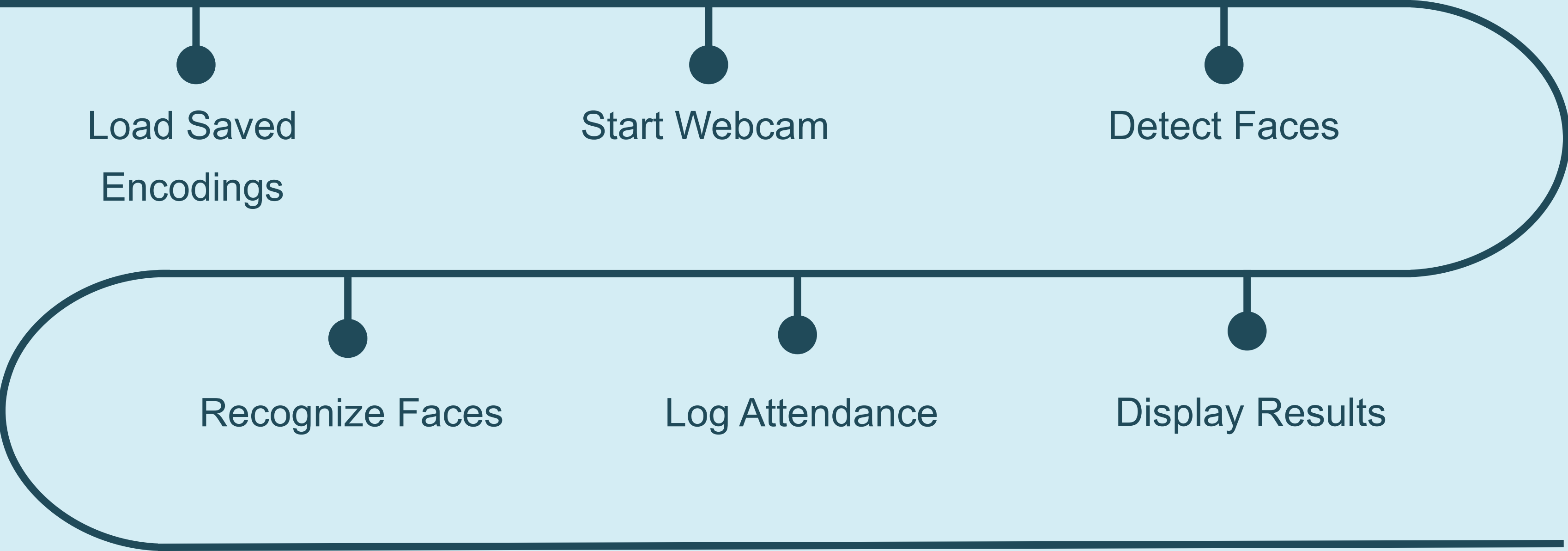
Load Known
Faces

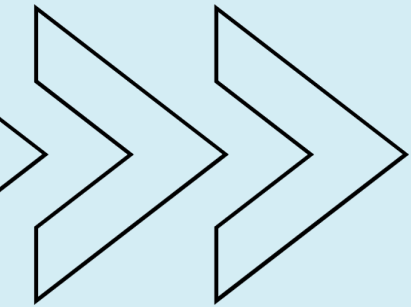
Encode Faces

Save Encodings

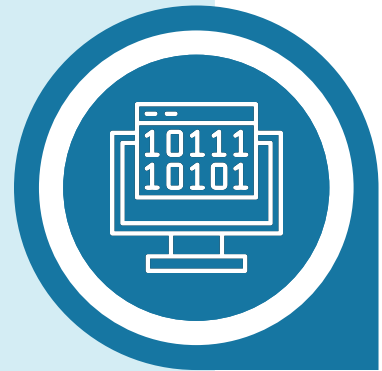
System Overview

- Runtime (Face Recognition)





Face Encoding

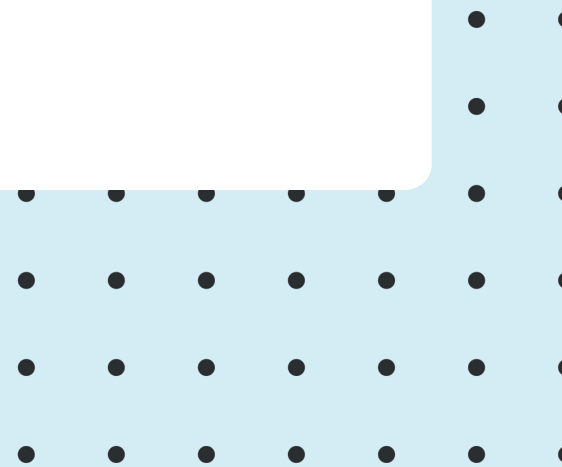


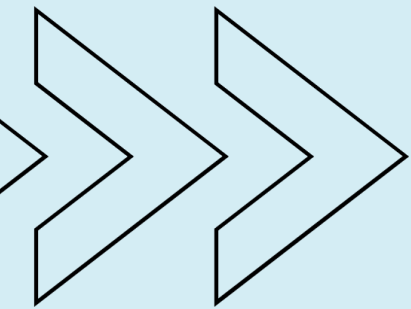
Encoding

Convert student face images into encodings and save them for future recognition.

Steps

1. Load images from `known_faces_data`
2. Detect faces in each image
3. Extract 128-d face encodings
4. Save all encodings to `encodings.pkl`





Face Recognition



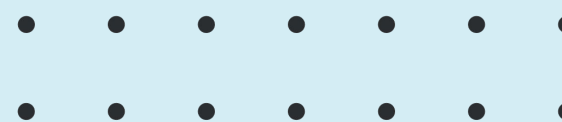
Recognition

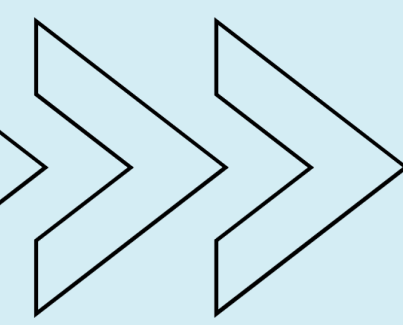
In real-time

Detect and recognize faces from the webcam.

Steps

1. Load saved face encodings (encodings.pkl)
2. Start webcam and capture video frames
3. Detect faces in each frame
4. Extract encoding for each detected face
5. Compare with known encodings
6. Recognize known students or label as "Unknown"





Attendance Logging

1. CSV Logging

Save student ID and avoid duplicates during session

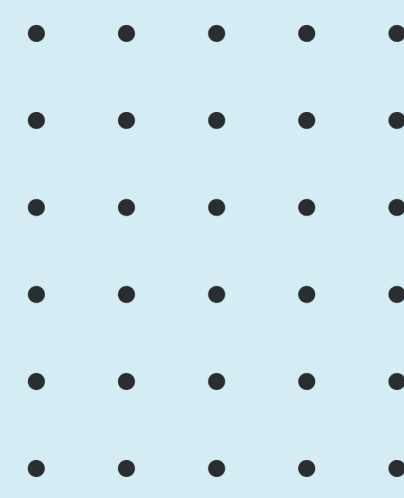
ID	Timestamp
208120250001	12:04:03
208120250002	12:04:10

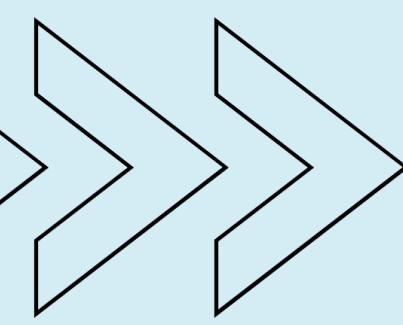
2. API Integration

Send GET request to university server with: Hall ID + Student ID

Endpoints Used:

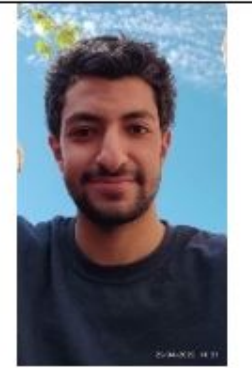
[https://.../api/attendance/{hall_id}](https://.../api/attendance/{hall_id}/{student_id})
[{student_id}](https://.../api/attendance/{hall_id}/{student_id})





Visual Results

Known Faces Data

		
Raafat	Abaza	Amr
		
Riyad	Sameh	Selim
		
Gasser	Omar	Tharwat

Results



-
-
-
-
-
-